

Comparing Real and Synthetic Image Representations in Stable Diffusion Models

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Abstract—Recent advancements in diffusion models have led to powerful generative frameworks capable of producing photorealistic images and rich latent representations. Despite being trained primarily to predict noise in a denoising process, these models leverage large-scale datasets, huge amounts of parameters, and attention mechanisms — resulting in emergent properties in their latent spaces. However, we lack a comprehensive understanding of the latent space of these models. For instance, in this paper, we investigate whether these models may be able to produce sufficiently distinguishable encodings for real and generated images in their latent space. We hypothesize that certain layers project these two image categories into distinct regions of the latent space, effectively separating real objects from synthetically generated ones. To verify this, we use real images sourced from two well-known datasets (COCO and Pascal VOC) and synthetic images generated by DALL-E 3, Midjourney, and Stable Diffusion (versions 1.4 and 1.5). We employ semantic segmentation (automated and manually verified) to isolate objects within each image and compute “prototypes” of their latent features. Our analysis reveals that, while most layers exhibit some level of separability between real and generated data, the 16×16 resolution layer of the decoder stands out as consistently enhancing the difference in latent space embeddings. These observations highlight the potential of using internal encodings of diffusion models not only for generative tasks but also for advanced analysis and recognition of real and synthetic visual content.

Index Terms—Stable Diffusion, internal representation, real-fake separability

I. INTRODUCTION

Stable Diffusion Models (SDMs) [1] have recently gained prominence due to their ability to generate high-fidelity images that rival human-made artwork and photography. This success comes largely from training on large image-text datasets, using large amounts of parameters, and employing attention mechanisms. Despite being originally designed to denoise corrupted images through a Markov chain, these models exhibit *emergent properties* in their internal representations — echoing similar phenomena observed in large language models [2].

Beyond image generation, diffusion models have demonstrated impressive capabilities for downstream applications. Using their pre-trained features for tasks such as semantic segmentation [3], fake detection [4], and other vision-based problems — mitigating the high computational costs of training these large models from scratch — proved to be highly effective. Following this growing interest in the internal representation produced by pre-trained SDMs, we propose a systematic approach to investigate the internal feature representation of *real* and *AI-generated* images. Understanding if the two kinds of images were encoded differently could have far-reaching implications: from detecting synthetic content and data filtering to strengthening defenses against adversarial attacks.

In this work, we explore the latent encoding distributions of real and generated images produced by an SDM. More concretely, we investigate whether images from real datasets such as COCO [5] and Pascal-VOC [6] are distinguishable from images generated by DALL-E 3 [7], Midjourney [8], and different versions of Stable Diffusion (v1.4 and v1.5) [1]. We explore various layers of the diffusion model’s UNet architecture, examining the difference between the internal representation of real and generated images. Following [9] and [10], we extract a “prototype” vector of each object in real and AI-generated images and compare them to evaluate whether there is a difference between the encoding of the two types of data (see Figure 1). Our empirical analysis indicates that the features extracted by the 16×16 layer of the decoder more clearly differentiate real and synthetic images in the latent space. The representation produced by this layer is further analyzed, at a dataset level, to better understand whether these representational differences can be attributed to background elements or foreground objects.

To the best of our knowledge, this is the first work to experimentally demonstrate that specific layers in an SDM produce different embeddings from real and synthetically

generated images.

This insight refines our understanding of how diffusion models organize semantic information internally and can serve as a stepping-stone for future research on content verification or detection strategies leveraging inherent properties of diffusion-based feature spaces. The remainder of this paper is organized as follows: Section II reviews relevant literature on diffusion models, while Section III details our methodology, including dataset selection and preprocessing. Section IV describes the experimental setup, while Section V presents and discusses the obtained results. Finally, Section VI concludes the paper with discussions on our findings and future directions.

II. RELATED WORK

A. Evolution of Diffusion Models

First introduced by Sohl-Dickstein et al. in [11], Diffusion Models (DMs) have represented a breakthrough in generative AI, as they have surpassed earlier generative approaches in both image quality and diversity, producing outputs that can be nearly indistinguishable from real images. Their success stems from a unique training approach that learns to reverse a gradual noising process, allowing for more stable training compared to adversarial methods [12]. In particular, DMs rely on a forward process that progressively adds noise to images and a reverse process guided by a U-shaped neural network (UNet), which systematically removes the noise to produce high-fidelity images from random noise.

Early implementations, such as the Denoising Diffusion Probabilistic Models (DDPMs) by Ho et al. [13], implemented both the forward and reverse processes as Markov chains. While effective, this approach involved calculating the distribution at each step, which made the computational cost nearly impractical for broader applications. To overcome this limitation, Denoising Diffusion Implicit Models (DDIMs) [14] introduced a non-Markovian framework, significantly reducing computational demands and enabling the generation of higher-resolution images. Nevertheless, both DDPMs and DDIMs operate directly in the high-dimensional pixel space, resulting in much higher computational costs compared to alternative generative models like Generative Adversarial Networks (GANs) and Variational Auto Encoders (VAEs). A breakthrough came with Latent Diffusion Models (LDMs) [15], which used a VAE to encode images into a lower-dimensional latent space. In this architecture, the diffusion process occurs in the latent space using a UNet, after which the VAE decoder converts the processed vector back to pixel space. This approach not only enhanced computational efficiency and reduced resource requirements but also enabled new capabilities, such as generating images conditioned on additional inputs [16], e.g. textual prompts or reference images. In this work, we aim to explore a pre-trained diffusion model known as SDM [1], a type of text-to-image LDM that has emerged as the state-of-the-art in image generation due to its capability to produce photorealistic images based on any given textual prompt.

B. Exploration and use of the DMs internal representation

To achieve state-of-the-art performance, DMs typically rely on extremely large datasets containing hundreds of millions [17] or even billions of examples [18]. The unaffordable scale of data and computational resources required to train a DM often make re-training these models from scratch infeasible. This limitation has encouraged the development of new research fields focused on exploring *pre-trained diffusion models* to uncover properties within their internal representations that can be leveraged to enhance their generative capabilities (for instance through textual prompt engineering [19]) or to tackle downstream tasks, including image classification and segmentation, all without the need for re-training. In [3], it is demonstrated that applying K-Means clustering to DDPM features produces a spatially coherent representation useful for downstream tasks, particularly semantic segmentation. They analyze different U-Net decoder layers across diffusion timesteps to identify the most effective for segmentation, then use its embedding vectors to train an MLP for pixel-wise labeling. In [20], the authors demonstrate that diffusion models can surpass GANs in image classification by leveraging their internal feature representations. Instead of using them solely for generation, the study extracts and pools discriminative features from the U-Net backbone of diffusion models to train a classifier. In ASYRP paper [21], the authors explore the internal representations of pretrained diffusion models, and introduce a semantic latent space termed “*h*-space”. This space exhibits homogeneity, linearity, robustness, and consistency across timesteps, making it suitable for straightforward image manipulation. In [22], the authors introduce LD-ZNet, a novel method for text-based segmentation of both real and AI-generated images by leveraging the latent space of LDMs. The key innovation lies in using the latent representations (*z*-space) learned by LDMs as input features for segmentation tasks, instead of traditional inputs like RGB images or CLIP embeddings. LD-ZNet significantly improves segmentation performance compared to traditional methods. In [23], a neural network is trained to modify the features of a pre-trained SDM to follow specific multimodal conditioning and improve generated image quality. The proposed model, ControlNet, adds spatial conditioning (e.g., edges, depth maps, segmentation maps, poses) to pretrained text-to-image diffusion models. By freezing the original model’s parameters and adding trainable layers, ControlNet enables precise control over image generation while maintaining stability during fine-tuning. More related to this work, in [9], [10] the authors extract the prototypes of objects belonging to different classes in various datasets to explore the capability of different layers of an SDM to preserve semantics.

The latent space of DMs has also been explored in the context of synthetic image detection and attribution. Zhang et al. [24] introduced the DE-FAKE framework, which leverages latent generative fingerprints to detect and attribute synthetic images to their respective source models. LDFaceNet [25] enhances fake image generation of faces, by leveraging the

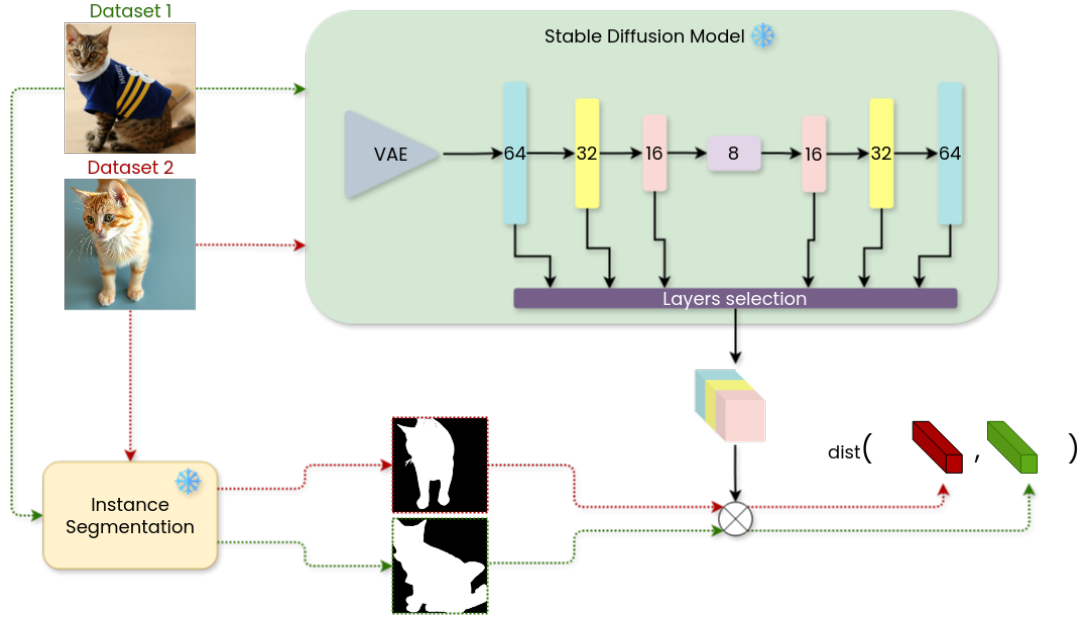


Fig. 1: **Overview of the framework employed for analyzing the internal representation of the SDM.** Given an image, corresponding SDM internal features are extracted from different layers and filtered based on the segmentation masks of the instances corresponding to the object. The filtered features are averaged and the prototypes of the objects thereby obtained are compared in terms of distance within the feature space. This allows us to compare the representation of object instances of real and synthetic images.

latent space of a DM, where a target image is encoded and fused with a specialized guidance vector to achieve more effective facial feature blending. FakeInversion [4], detects fake images by leveraging the internal features of a pre-trained SDM. The authors train a ResNet-50 classifier on features extracted from the diffusion model’s latent space, rather than relying on raw pixel data. By inverting images into this feature space, the classifier learns to distinguish real from synthetic images, enabling generalization to unseen generative models. *Unlike prior works that rely on external classifiers or post-hoc analysis, this paper examines the potential of SDM to create different representations for real and synthetic images within its latent space.*

III. METHODS

The procedure used to extract the object prototypes from each image is described in Section III-A, whereas the four datasets employed in this study are presented in Section III-B. Section III-C details the data pre-processing steps for generating instance segmentation maps of AI-generated images and selecting the subset of real and synthetic images.

A. Prototype Extraction

The procedure employed in this study to analyze the internal features of an SDM can be summarized as follows (see Figure 1). For each input image, the inference at $t = 0$, corresponding to the final denoising step in the stable diffusion UNet, is performed. During this process, the image is first encoded by the VAE, yielding a latent representation that is

then fed into the UNet. Feature maps are extracted from the selected residual blocks of UNet and instance segmentation masks are applied within the feature space to extract the representation of specific objects. Each mask is rescaled to match the feature map resolution, effectively filtering out all but the features corresponding to the target object. Finally, an object-level descriptor is obtained by averaging the masked region across spatial dimensions, resulting in a prototype vector that encapsulates the characteristic features of each object instance. By calculating the distance between prototypes, it is possible to measure the distance between objects in the embedding space. The described procedure is inspired by [9] and it is used to compare embeddings of real and generated images.

B. Datasets

To compare the internal representation of real and AI-generated images, in our study, we employ four datasets: two containing real images and two containing synthetically generated images.

(a) Real Image Datasets:

- Pascal-VOC [6] is a widely used benchmark for image segmentation. Its images feature a limited set of object categories, typically with the primary object centered in the scene. The dataset includes 20 object categories plus a background class and a “do not care” class for uncertain regions. Both training and validation sets provide pixel-level annotations.
- COCO [5] is a comprehensive dataset containing over 100,000 images designed for object detection, segmen-

tation, and captioning tasks. Unlike Pascal-VOC, COCO images often capture complex scenes with multiple objects. The dataset provides instance-level annotations for 80 object categories and background labels across its training and validation sets.

(b) Synthetic Image Datasets:

- GenImage [26] is a large-scale dataset specifically created for detecting AI-generated images. It comprises approximately 1.35 million synthetic images generated by eight different AI models: BigGAN [27], GLIDE [28], VQDM [29], Stable Diffusion (v1.4 and v1.5) [1], ADM [12], Midjourney [8], and Wukong [30]. The dataset includes the 1,000 ImageNet [31] classes, with each class containing roughly 1,350 fake images (1,300 for training, and 50 for testing). Images were generated using simple text prompts following the template “photo of [class]” and feature resolutions ranging from 128×128 to 1024×1024 , depending on the generator model.
- ImagiNet [32] contains 200,000 high-resolution images balanced between real and synthetic content. The synthetic images are generated using various methods including GANs [33], diffusion models [13], and commercial tools like Midjourney [8] and DALL-E 3 [7], while real images are collected from public datasets. The dataset supports two evaluation tasks: distinguishing between real and synthetic images and identifying the specific generative model used for generation.

C. Data Pre-Processing

Following the procedure described in Section III-A, we examine the features extracted from the four datasets described in Section III-B. Since GenImage and ImagiNet lack instance segmentation labels, we employ a pre-trained segmentation network to generate the required instance segmentation maps. We utilize Mask2Former [34], a universal image segmentation architecture capable of handling multiple segmentation tasks (panoptic, instance, and semantic) within a single framework. The model comprises a feature extractor, a pixel decoder, and a transformer decoder, with its key innovation being masked attention that restricts attention to localized features around predicted segments. We specifically use the large-sized version of Mask2Former pre-trained on COCO instance segmentation with a swin backbone [35], implemented by Hugging Face¹. To ensure the goodness of the segmentation masks, the predicted label maps are visually inspected to remove those with inaccurate segmentation results. While GenImage and ImagiNet contain images from various generators, we focus our study on outputs from four specific models: Midjourney, Stable Diffusion v1.4, Stable Diffusion v1.5 (from GenImage), and DALLE3 (from ImagiNet). In our analysis, we consider eight classes common across all datasets: background, cat, dog, person, car, bird, boat, and train. To ensure meaningful analysis, we select only images where objects from these classes occupy more than 10% of the image area, ensuring the

objects are visible and well-represented. This selection criteria yield a total of 4,037 images, the number of images in each dataset is reported in Table I:

Dataset	Number of Images
Pascal-VOC	654
COCO	1,000
DALLE3	709
Stable Diffusion 1.5	547
Stable Diffusion 1.4	566
Midjourney	561

TABLE I: Number of images in each dataset.

IV. EXPERIMENTAL SETUP

To analyze the features extracted from the diffusion model’s internal layers, we design an experimental framework using prototypes obtained from both real and AI-generated images, following the extraction procedure detailed in Section III-A. Our experiments utilize a pre-trained Stable Diffusion Model (v1.5), through its publicly available implementation², as features extractor. All images are resized to 512×512 pixels before processing.

To compare object prototypes across datasets, we employ the Distance Correlation [36], which for two n-dimensional vectors $\mathbf{X} = (x_1, x_2, \dots, x_n)$ and $\mathbf{Y} = (y_1, y_2, \dots, y_n)$ is defined as:

$$D_{\text{corr}}(\mathbf{X}, \mathbf{Y}) = \sqrt{\frac{dCov^2(\mathbf{X}, \mathbf{Y})}{\sqrt{dVar^2(\mathbf{X})dVar^2(\mathbf{Y})}}},$$

where the distance covariance $dCov^2(\mathbf{X}, \mathbf{Y})$ and the distance variance $dVar^2(\mathbf{X})$ between $\mathbf{X}$ and $\mathbf{Y}$ are

$$dVar^2(\mathbf{X}) = dCov^2(\mathbf{X}, \mathbf{X})$$

$$dCov^2(\mathbf{X}, \mathbf{Y}) = \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n (a_{ij} - \bar{a}_{i.} - \bar{a}_{.j} + \bar{a})(b_{ij} - \bar{b}_{i.} - \bar{b}_{.j} + \bar{b})$$

If x_i, x_j are two samples of $\mathbf{X}$ and y_k, y_l are two samples of $\mathbf{Y}$, it is possible to define:

$$a_{ij} = D_{\text{euc}}(x_i, x_j), \quad b_{kl} = D_{\text{euc}}(y_k, y_l), \quad \bar{a}_{i.} = \frac{1}{n} \sum_{j=1}^n a_{ij},$$

$$\bar{a}_{.j} = \frac{1}{n} \sum_{i=1}^n a_{ij}, \quad \bar{b}_{i.} = \frac{1}{n} \sum_{j=1}^n b_{ij}, \quad \bar{b}_{.j} = \frac{1}{n} \sum_{i=1}^n b_{ij}$$

Our primary objective is to investigate the features extracted from different diffusion UNet layers to study their ability to effectively distinguish between real and AI-generated images. We focus our attention on layers in both the encoder and decoder with spatial resolutions of 32×32 and 16×16 , extracting object prototypes from each layer. To evaluate how effectively different layers capture distinctive features between real and synthetic images, we compute four sets of distances across our datasets:

¹<https://huggingface.co/facebook/mask2former-swin-large-coco-instance>

²<https://github.com/hkproj/pytorch-stable-diffusion>

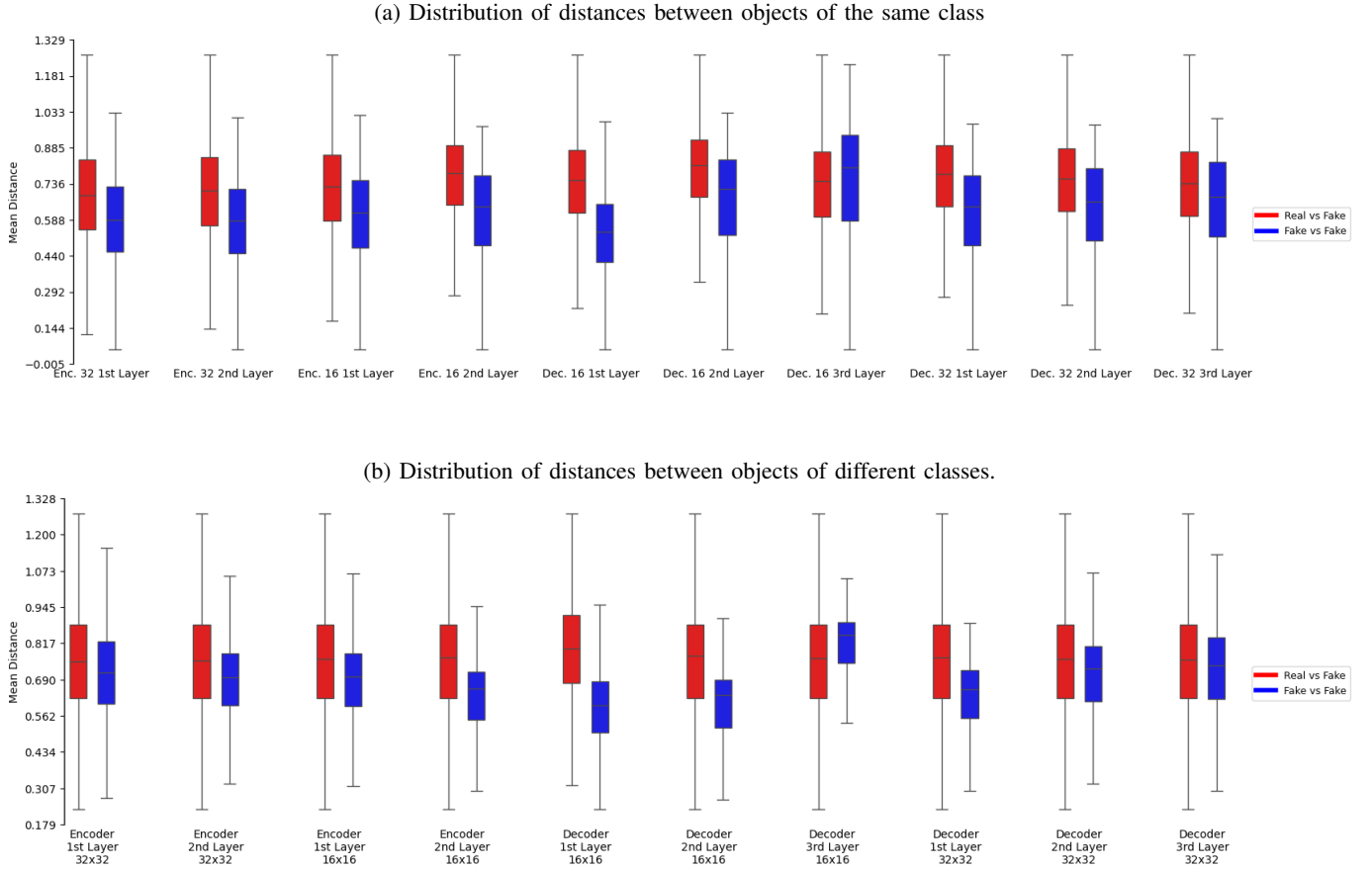


Fig. 2: The box plots compare the distributions of feature distances between real and generated images, extracted from different layers of the diffusion UNet. Panel (a) illustrates the feature distances computed between objects belonging to the same class, while panel (b) depicts feature distances between objects from different classes.

- **Real vs Fake (same classes)** – Distances between prototypes from real datasets (Pascal-VOC or COCO) and their corresponding class prototypes from AI-generated images (DALL-E 3, Midjourney, Stable Diffusion v1.4 and v1.5).
- **Real vs Fake (different classes)** – Distances between prototypes of different classes, comparing real images against synthetic ones.
- **Fake vs Fake (same classes)** – Distances between same-class prototypes within AI-generated images, specifically comparing the same object classes generated by the same AI model.
- **Fake vs Fake (different classes)** – Distances between different-class prototypes within AI-generated images, specifically comparing different object classes generated by the same AI model.

A layer’s effectiveness in distinguishing between real and synthetic images can be assessed by measuring the separation between two distance distributions: the Real vs Fake distances and the Fake vs Fake distances. An ideal discriminative layer would show a clear separation between these distributions, both when comparing same-class objects and when comparing

objects across different classes. This separation would indicate that the layer is capable of generating features that consistently differentiate between real and AI-generated content, regardless of the object classes being analyzed.

Building on this initial finding, we conduct a more granular analysis of the layer that shows promising discriminative capabilities between real and synthetic images. Rather than aggregating distances across all the datasets, we examine each dataset’s relationships independently. This dataset-specific approach allows us to map out the similarities between features of objects generated by the same tool, compare them against features from different generators, and contrast them with features from real images. For example, taking Midjourney-generated images as a reference point, we analyze five distinct distance sets:

- Midjourney vs COCO
- Midjourney vs Pascal-VOC
- Midjourney vs DALL-E 3
- Midjourney vs Stable Diffusion v1.4
- Midjourney vs Stable Diffusion v1.5

We repeat this comparative analysis for images generated by DALL-E 3, Stable Diffusion v1.4, and Stable Diffusion v1.5,

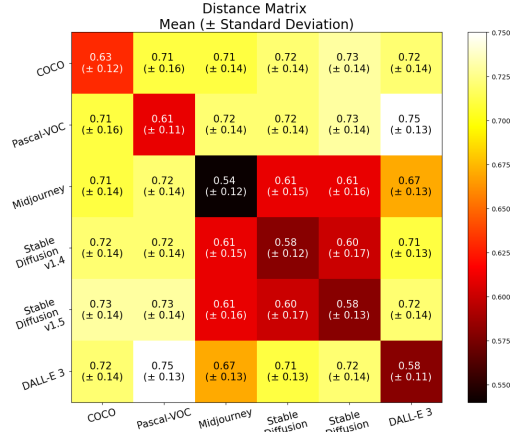
computing distances both within and across object classes. In an ideal scenario, we would observe a clear separation between these distance sets, with features from objects created by the same generator showing greater similarity to each other than to either real objects or those produced by other AI tools. Furthermore, UMAP (Uniform Manifold Approximation and Projection) for Dimension Reduction [37] is used to plot the distribution of features extracted from real and fake data. Using the UMAP visualizations, we also investigate how the background is encoded by plotting its features separately from those of the other analyzed objects. This detailed analysis highlights how different generative models encode objects in distinctive ways, suggesting the potential to leverage these patterns as identifying “fingerprints” potentially useful for attribution or detection tasks.

V. RESULTS

Following the experimental setup detailed in Section IV, we conduct an analysis of different layers of the diffusion UNet to effectively extract distinct features from real and AI-generated images. Figure 2 presents box plot distributions of feature distances between real and synthetic images, revealing how each network layer characterizes the differences between authentic and generated data. Our analysis demonstrates the progressive evolution of feature representations through the network, with distinct patterns emerging at different depths. In the box plots, the distributions are captured by their median values and interquartile ranges, highlighting how effectively each layer’s representation distinguishes between real and synthetic images. Specifically, this layer-wise comparison enables us to identify which network depths are most sensitive to the subtle differences between real and synthetic images. The analysis shows that the first layer of the UNet decoder at 16×16 resolution (Decoder, 1st Layer, 16×16) demonstrates particularly strong discriminative ability in separating real from synthetic images. This capability is evident in both experimental scenarios — when comparing features from objects of the same class (Figure 2a) and across different classes (Figure 2b). Indeed, the distance distributions of real and synthetic images show a minimal overlap, with only the tail regions intersecting. This separation in feature space suggests that the layer effectively captures distinctive signals, making it particularly adept at differentiating real and generated images. To further investigate the discriminative capabilities of the layer, we conduct a detailed analysis of its feature characteristics across different datasets. Figure 3 presents two distance matrices displaying the mean and standard deviation of inter-dataset distances, comparing objects of the same class (Figure 3a) and across different classes (Figure 3b).

The diagonal elements of both matrices exhibit the lowest distance values, indicating strong feature similarity when comparing images from the same dataset, whether real or synthetic. Notably, the self-comparison distances within synthetic datasets are substantially lower than their distances to real images, as evidenced by the values in the first two rows (and columns) of the matrices. When examining synthetic images

(a) Mean distances between objects of the same class



(b) Mean distances between objects of different classes.

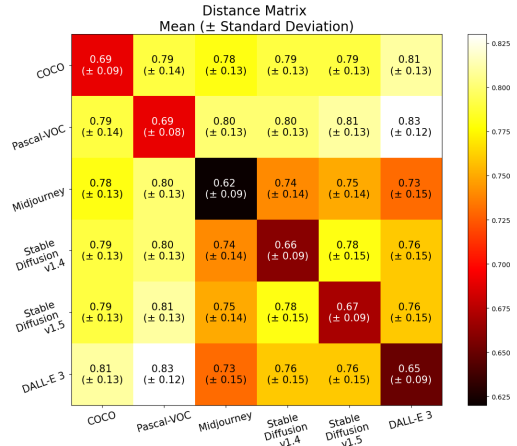


Fig. 3: Distance matrices that compare the objects’ features in different datasets.

cross-dataset relationships, we find that Midjourney and Stable Diffusion (v1.4 and v1.5) yield features that overlap considerably for the same object categories, making it challenging to distinguish which model produced a given instance. However, differences across object classes are more pronounced, implying that class categories heavily influence how these models organize their latent representation. Meanwhile, DALL-E 3 stands out from both real images and other synthetic sources, likely due to its consistently recognizable visual style, which remains evident across classes.

These results suggest that using features extracted from a specific layer of a pre-trained DM it is possible to reveal a fingerprint that characterizes images generated with a specific AI tool. To further validate this hypothesis, we conducted a dimensionality reduction analysis using UMAP. This technique allows to visualize the 1280-dimensional object prototypes in an interpretable 3D space. Figure 4 presents separate 3D plots

for each synthetic dataset, comparing their feature distributions against real images while distinguishing between background and foreground objects. Despite the inherent information loss in dimensionality reduction, the 3D visualizations reveal clear structural patterns in the feature space. Across all datasets, features are consistently organized into two primary clusters corresponding to background and foreground objects. Within these primary clusters, a finer substructure emerges: the features of real and synthetic images form distinct sub-clusters, indicating systematic differences in their representations. This separation is particularly pronounced for foreground objects, where synthetic images consistently exhibit feature distributions that deviate notably from their real counterparts across all generative models examined.

VI. CONCLUSION AND FUTURE WORK

The results presented in this paper indicate that certain layers of a Stable Diffusion UNet encode distinct representations for real and synthetic images. By applying a segmentation-guided prototype extraction framework, it was shown that the SDM produces different representations of real and synthetic data. This behavior is particularly evident in the first layer of the decoder operating at a 16×16 spatial resolution, where the latent representations of real and synthetic objects exhibit minimal overlap.

An in-depth dataset-specific comparison reveal that distance matrices systematically produce lower intra-dataset distances than inter-dataset ones, strengthening the hypothesis that synthetic images from the same generator share a characteristic “fingerprint” in the latent space. Notably, features generated by DALL-E 3 differ substantially from those produced by Stable Diffusion or Midjourney, underscoring that each generator creates a distinctive embedding. A dimensionality reduction analysis using UMAP further showed coherent clustering according to both background and foreground content, while also distinguishing real from synthetic data within each cluster.

Taken together, these findings suggest that the first layer of the decoder at the spatial resolution of 16×16 in a pre-trained diffusion model can robustly encode discriminative features that differentiate real from AI-generated content. This raises the potential for leveraging internal representations of large-scale generative models not only in few-shot or zero-shot approaches for segmentation or classification but also to develop targeted fake image recognition strategies. Future work will aim to integrate such insights into efficient image fake detection tools and investigate whether these detected differences hold across an even broader range of generative methods and image categories.

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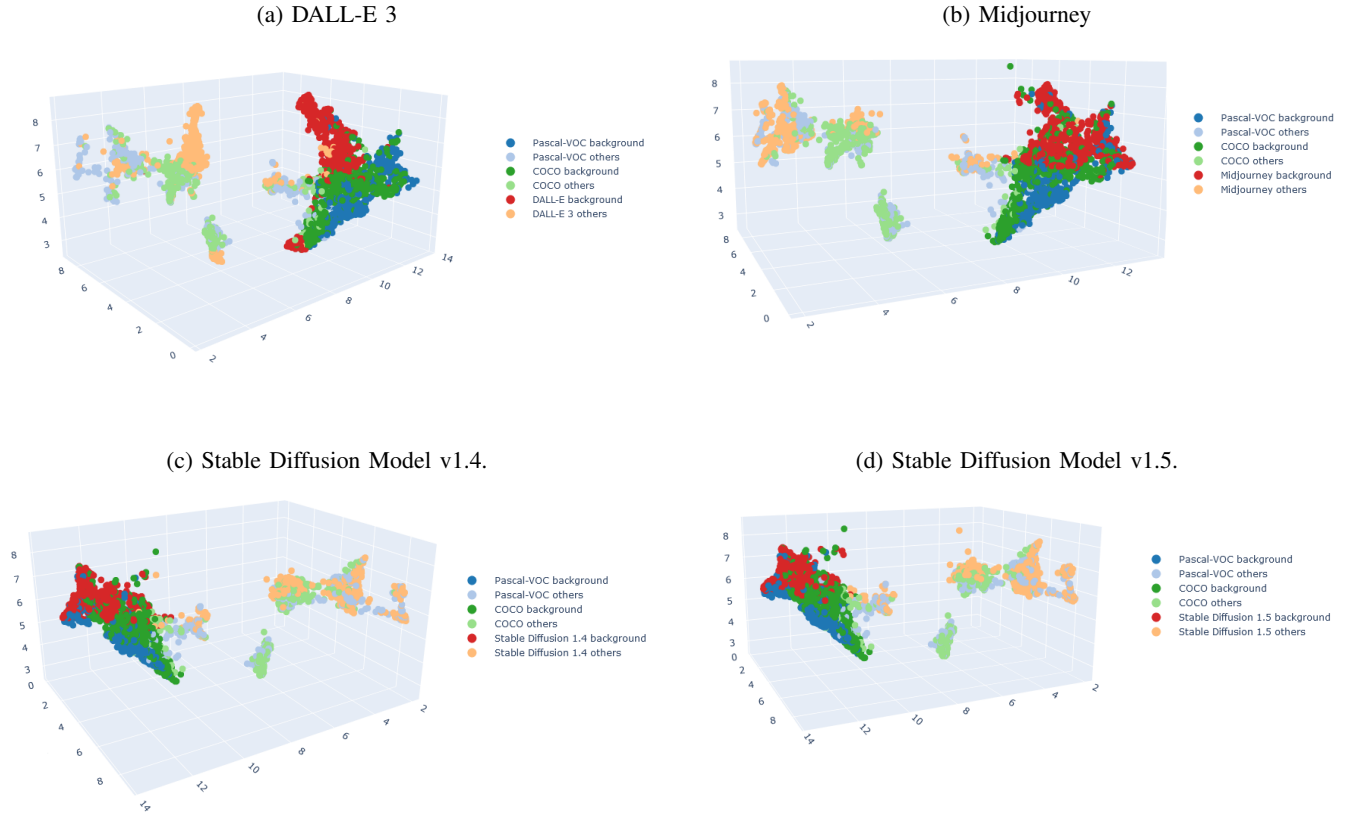


Fig. 4: 3D feature distributions of real and generated objects obtained using UMAP.

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